DATA SCIENCE CAPSTONE PROJECT

APPLIED DATA SCIENCE CAPSTONE

Predicting SpaceX Falcon 9 first-stage booster reuse outcomes using a custom REST API pipeline, web scraping, geospatial analytics, interactive dashboards, and supervised classification machine learning models.

Lahcen Mokhtar AMROUNE | July 2026

https://github.com/LahcenAmroune/Applied-Data-Science-Capstone

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Executive Summary & Overview

This project evaluates the predictability of SpaceX Falcon 9 first-stage booster landings to analyze commercial launch economics and competitive pricing dynamics. By successfully recovering first-stage boosters—which represent over 60% of total rocket production costs—SpaceX offers launch prices near $62M, drastically undercutting traditional industry benchmarks of $165M+.

Using an end-to-end data pipeline combining REST API collection, web scraping, exploratory data analysis (EDA), and interactive geospatial dashboards, we examined how payload mass, orbit selection, and launch pad geography influence booster recovery. Machine learning models were benchmarked to predict landing success, with the **Decision Tree classifier emerging as the top-performing algorithm**. These predictive capabilities enable competitors and launch customers to assess mission recovery risk and optimize commercial bidding strategies.

Introduction

This capstone project aims to develop a predictive model capable of determining whether the first stage of a SpaceX Falcon 9 rocket will land successfully.

SpaceX currently advertises Falcon 9 launches at a cost of $62 million, significantly undercutting competitors who charge upwards of $165 million per launch. This substantial cost advantage is primarily driven by SpaceX's ability to recover and reuse the first-stage booster. Accurately predicting landing outcomes therefore provides critical insight into launch costs, enabling potential competitors to better assess pricing strategies when bidding against SpaceX.

It should be noted that not all unsuccessful landings are failures in the operational sense; some are intentional, such as controlled oceanic landings performed for mission-specific reasons.

This project seeks to build a predictive model that uses launch parameters like payload mass, orbit type, and site location to forecast whether the Falcon 9 booster will successfully touch down.

Methodology

- Data collection methodology :
  - Retrieval and consolidation from multiple SpaceX API endponts
  - Web scaping tabular data from Wikipedia

- Perform data wrangling
  - Extracted relevant records
  - Flattened fields and resolved missing values endpoints

- Perform exploratory data analysis (EDA) using visualization and SQL
  - Visualize variable relationships
  - Look at the data in aggregate

- Perform interactive visual analytics using Folium and Plotly Dash
  - Mark all launch sites on a map
  - Mark successful and failed launches
  - Calculate distances to proximate locations
  - Provide for interactive exploration of the data
- Perform predictive analysis using classification models 7
  - Build, evaluate, and compare several predictive classification models

Data Collection

- Data was collected using a combination of retrieval techiniques:
  - HTTP requests, SpaceX API :
    - Initial lauch data was obtained from : /v4/launches/past
    - Additional data was backfilled from:
      - /V4/rockests
      - /V4/launchpads
      - /V4/playloads
      - /V4/cores
    - Tabular data from the : List of Falcon 9 and Falcon Heavy launches – Wikipedia page.

Data collection, wrangling and formating

The primary data source for this project was the SpaceX API, provided through the Coursera platform at https://api.spacexdata.com/v4/rockets/. We utilized this API to retrieve launch records; however, as our analysis focuses exclusively on Falcon 9 missions, the data was filtered accordingly to include only Falcon 9 launches.

During the data cleaning process, missing values were handled by imputing the mean of the respective columns. After preprocessing, the final cleaned dataset comprised 90 launch records (rows) and 17 features (columns). A preview of the dataset following cleaning is presented below.

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	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	Reu
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False	None	1.0	
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None	1.0	
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	

Data collection, wrangling and formating

In addition to the API data, we also collected information through web scraping. The source used was the Wikipedia page dedicated to Falcon 9 and Falcon Heavy launches, accessible at the following URL:
https://en.wikipedia.org/w/index.php?title=list_of_falcon_9_and_falcon_heavy_launches&oldid=1027686922

This page was specifically chosen because it contains data exclusively related to Falcon 9 launches, making it directly relevant to our analysis. After extracting and processing the scraped data, we obtained a cleaned dataset consisting of 121 launch records (rows) and 11 features (columns).

A sample of the first few entries from this dataset is shown in the table below.

Let me know if you need any adjustments!

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1	No attempt\n	1 March 2013	15:10

Data collection, wrangling and formating

This phase was structured around two main goals: uncovering patterns in the launch data, and preparing the target label for supervised learning.

To accomplish this, we followed a systematic process:

- Data Assessment – We began by checking for missing values, calculating the percentage of nulls per attribute, and classifying each column as either numerical or categorical.
- Distribution Analysis – We then examined the spread of launches across different launch sites and orbit types to understand the composition of the dataset.
- Outcome Preparation – Finally, we explored mission outcomes, grouped them into binary categories (success or failure), and stored the result as a "Class" label, which served as the target variable for training our models.

[33]:
```
df['Class']=landing_class
df[['Class']].head(8)
```

[33]:

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/1%20Data%20Collection%20API.ipynb

https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/2%20Data%20Collection%20with%20Web%20Scraping.ipynb

https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/3%20Data%20Collection%20data%20wrangling.ipynb

Exploratory Data Analysis (EDA)

Data Processing & Analysis

- Libraries: Pandas, NumPy, and SQL.
- Pandas/NumPy: Used for initial data exploration, generating summary statistics on launch sites, orbit types, and mission outcomes.
- SQL: Employed for targeted queries to extract specific insights, such as unique launch sites, total payload mass for NASA (CRS) missions, and average payload mass for the F9 v1.1 booster.

Data Visualization

- Libraries: Matplotlib, Seaborn, Folium, and Dash.
- Matplotlib/Seaborn: Create static charts (scatter, bar, line) to visualize relationships between flight numbers, payload mass, launch sites, and orbit success rates.
- Folium: Generates interactive maps to pinpoint launch sites, visualize mission outcomes (success/failure) per site, and measure distances to nearby infrastructure (cities, railways, highways).
- Dash: Powers an interactive dashboard featuring dropdowns and sliders, allowing users to explore total successful launches per site and the correlation between payload mass and mission outcome.

Exploratory Data Analysis (EDA)

Machine Learning Prediction

Library: Scikit-learn.

Process: The workflow includes data standardization and splitting data into training/testing sets.

Models: Implements Logistic Regression, SVM, Decision Trees, and KNN.

Evaluation: Models are tuned via hyperparameter optimization and assessed using accuracy scores and confusion matrices.

EDA with SQL

SQL Queries :

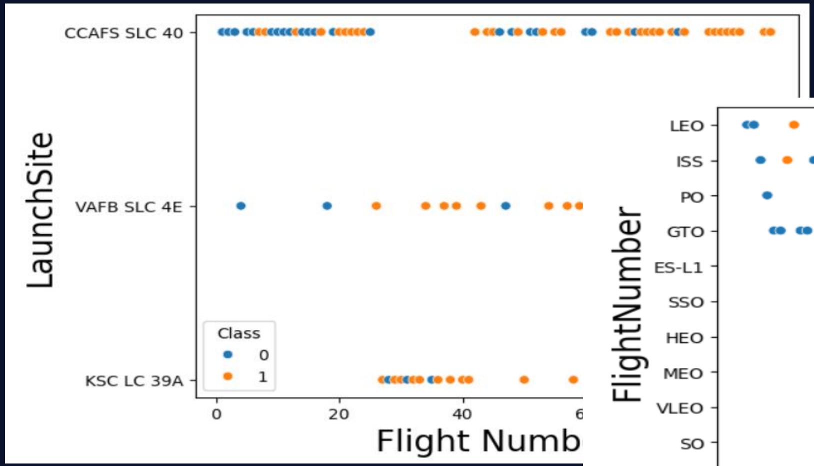
- Show each unique launch site
- Show 5 records where launch site names begin with 'CCA'
- Display the total payload mass carried by boosters launched by 'NASA (CRS)'
- Display the average payload mass carried by the v1.1 Falcon 9 booster
- List the date of the first successful ground landing outcome
- List the booster versions with successful outcomes landing on the drone ship with payloads between 4000kg and 6000kg.
- List the number of successful and failed mission outcomes
- List all of the booster versions that carried the max payload mass
- List the month name, outcome, booster version, and launch site for missions with failure outcomes landing on a drone ship in 2015.
- Show the distribution of outcomes between June 4th, 2010 and March 20th, 2017

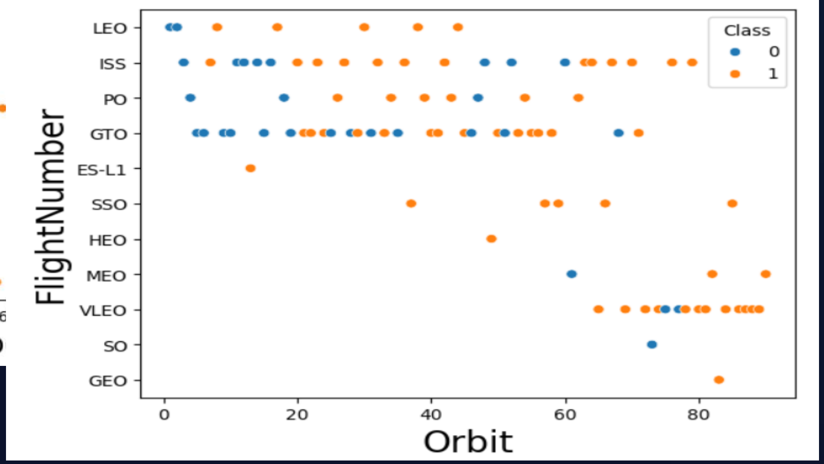
https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/4%20EDA%20SQL.ipynb

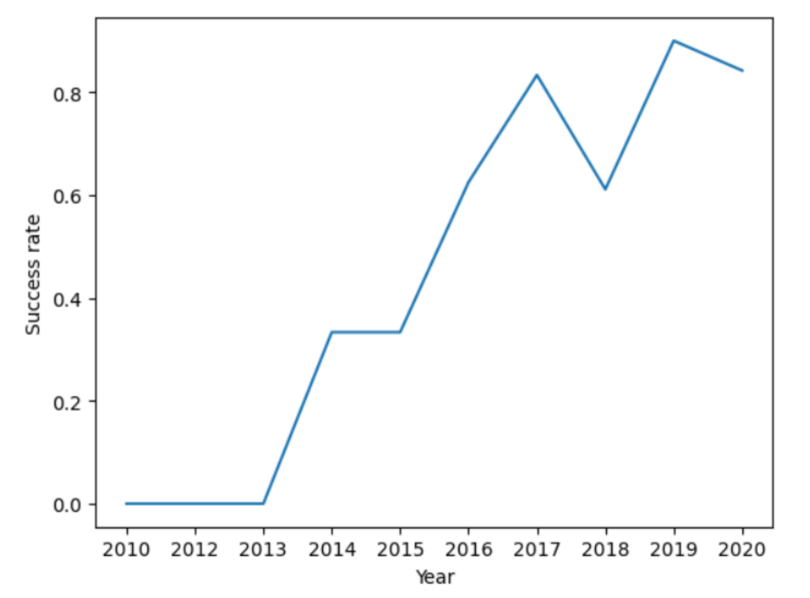
EDA with Data Visualization

Smploy data visualization techniques to evaluate the contribution of individual features and their impact on target outcomes.:

- Visualize the relationship between Flight Number and Launch Site
- Visualize the relationship between Payload and Launch Site
- Visualize the relationship between success rate of each orbit type
- Visualize the relationship between FlightNumber and Orbit type
- Visualize the relationship between Payload and Orbit type
- Visualize the launch success yearly trend







EDA with Data Visualization

- Features Engineering : Creation of dummy variables to categorical columns

	FlightNumber	PayloadMass	Flights	GridFins	Reused	Legs	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	...	Serial_B1048	Serial_B1049	Serial_B1050	Serial_B1051	Serial_B1054	Serial_B105
0	1	6104.959412	1	False	False	False	1.0	0	False	False	...	False	False	False	False	False	Fals
1	2	525.000000	1	False	False	False	1.0	0	False	False	...	False	False	False	False	False	Fals
2	3	677.000000	1	False	False	False	1.0	0	False	False	...	False	False	False	False	False	Fals
3	4	500.000000	1	False	False	False	1.0	0	False	False	...	False	False	False	False	False	Fals
4	5	3170.000000	1	False	False	False	1.0	0	False	False	...	False	False	False	False	False	Fals
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
85	86	15400.000000	2	True	True	True	5.0	2	False	False	...	False	False	False	False	False	Fals
86	87	15400.000000	3	True	True	True	5.0	2	False	False	...	False	False	False	False	False	Fals
87	88	15400.000000	6	True	True	True	5.0	5	False	False	...	False	False	False	True	False	Fals
88	89	15400.000000	3	True	True	True	5.0	2	False	False	...	False	False	False	False	False	Fals
89	90	3681.000000	1	True	False	True	5.0	0	False	False	...	False	False	False	False	False	Fals

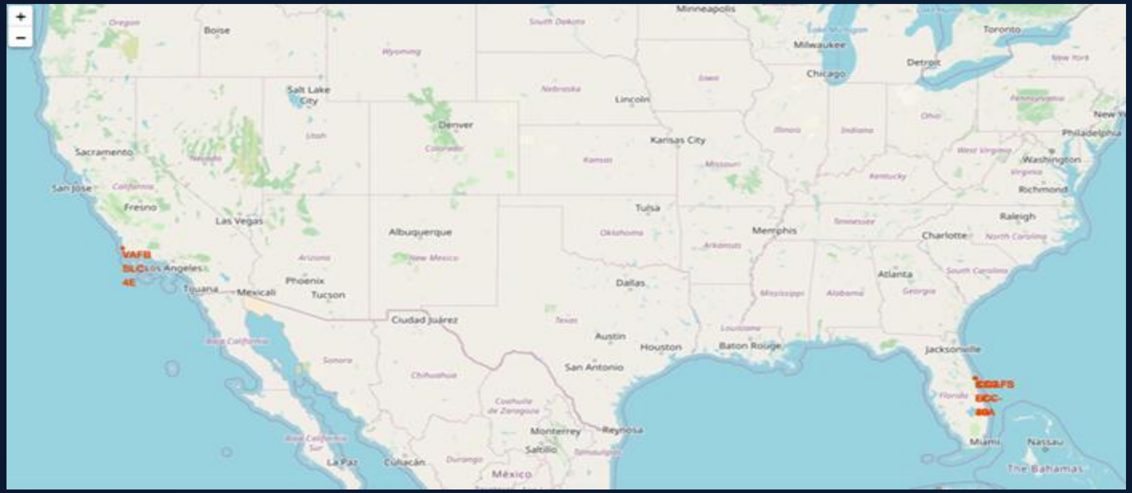
90 rows × 80 columns

https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/5%20EDA%20Visualization.ipynb

Interactive Visual Analytics with Folium

identify spatial patterns, the following geographic features were plotted on a map of launch sites:

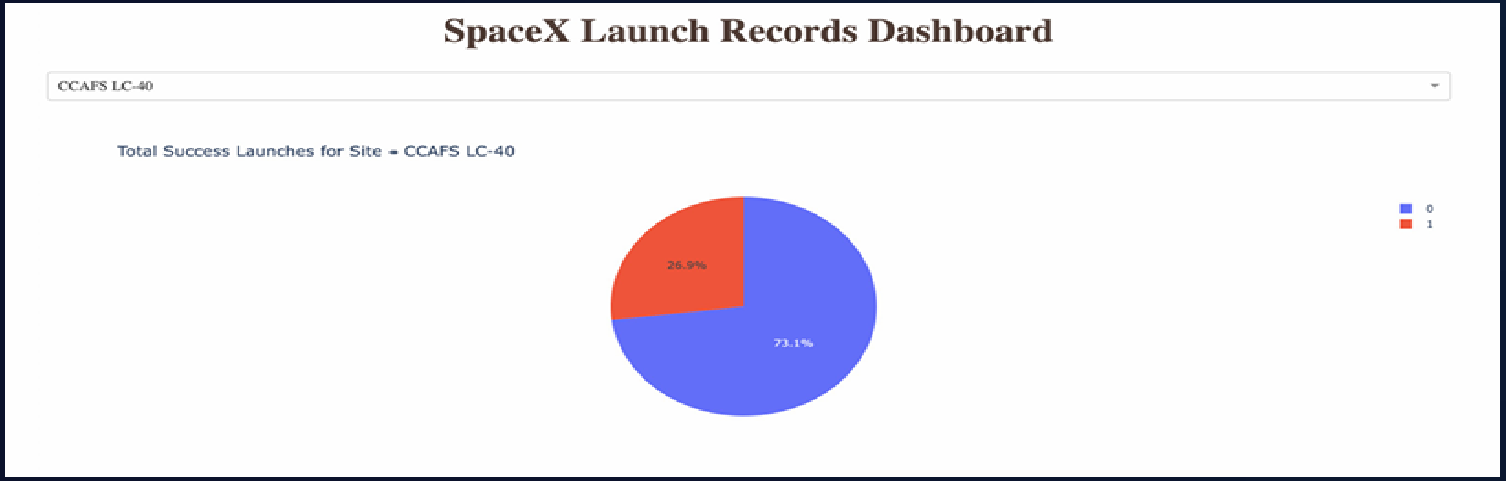
- All Launch Sites
- Successful and Failed Launches
- Distances between a launch site and proximate landmarks
- Geographic analysis was conducted by visualizing launch site coordinates, mission outcome distributions, and distances to key landmarks on an interactive map.



https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/6%20Interactive%20Visual%20Analytics%20with%20Folium%20lab.ipynb

Interactive Visual Analytics and Dashboard

- The figure below presents a pie chart illustrating the launch outcomes for site CCAFS LC-40, where 0 denotes failed launches and 1 denotes successful launches. As shown, 73.1% of launches from this site resulted in failure.
- The pie chart below visualizes the outcome distribution for CCAFS LC-40, with failure represented by 0 and success by 1. Notably, the majority of launches—73.1%—from this site were unsuccessful.



Interactive Visual Analytics and Dashboard

- • The picture below shows a scatterplot when the payload mass range is set to be from 2000kg to 8000kg.
- • Class 0 represents failed launches while class 1 represents successful launches.

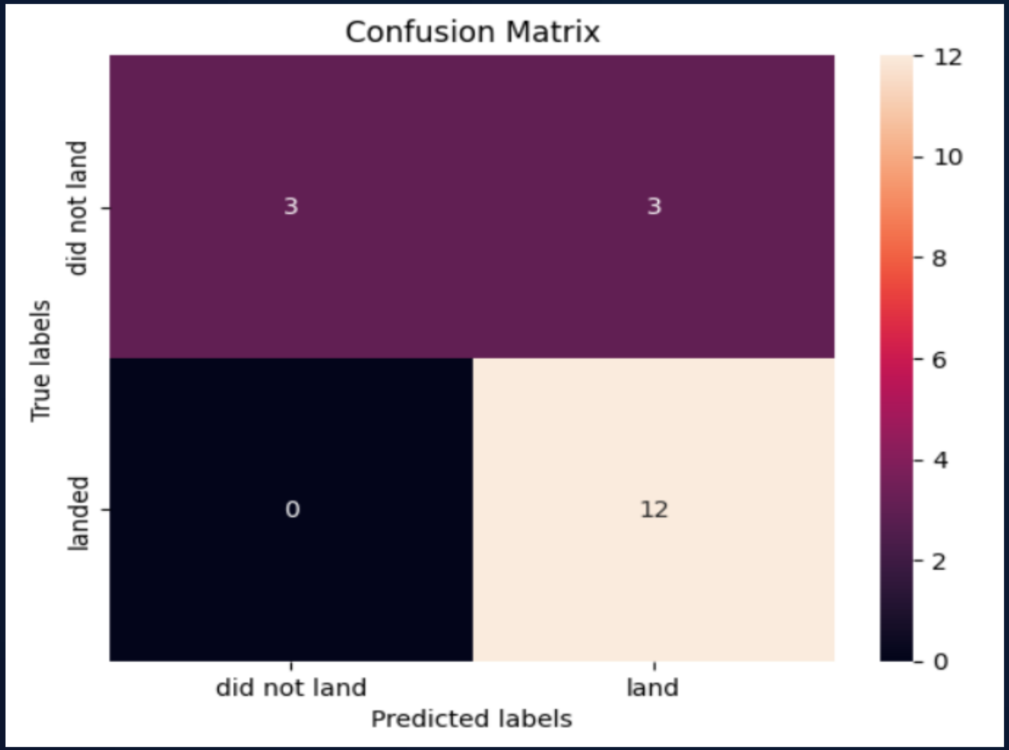
https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/7%20spacex%20dash%20app.py

Machine Learning Prediction

Logistic regression

- GridSearchCV best score: 0.8464285714285713
- Accuracy score on test set: 0.8333333333333334

Confusion matrix:

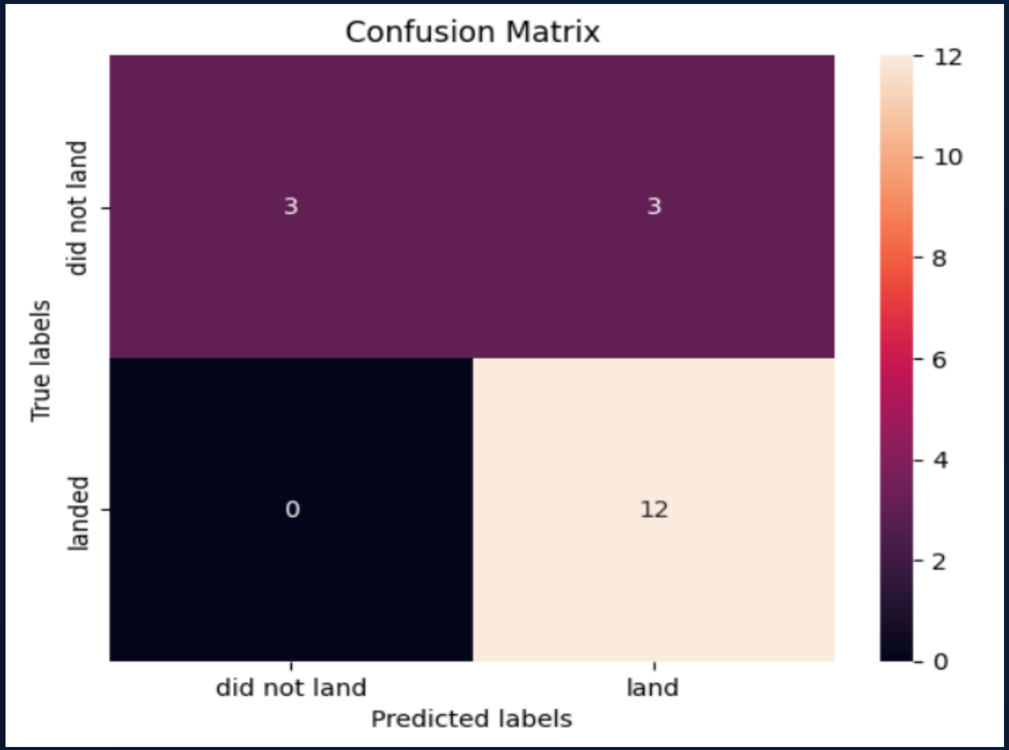


Machine Learning Prediction

Support Vector Machine (SVM)

- GridSearchCV best score: 0.8482142857142856
- Accuracy score on test set: 0.8333333333333334

Confusion matrix:

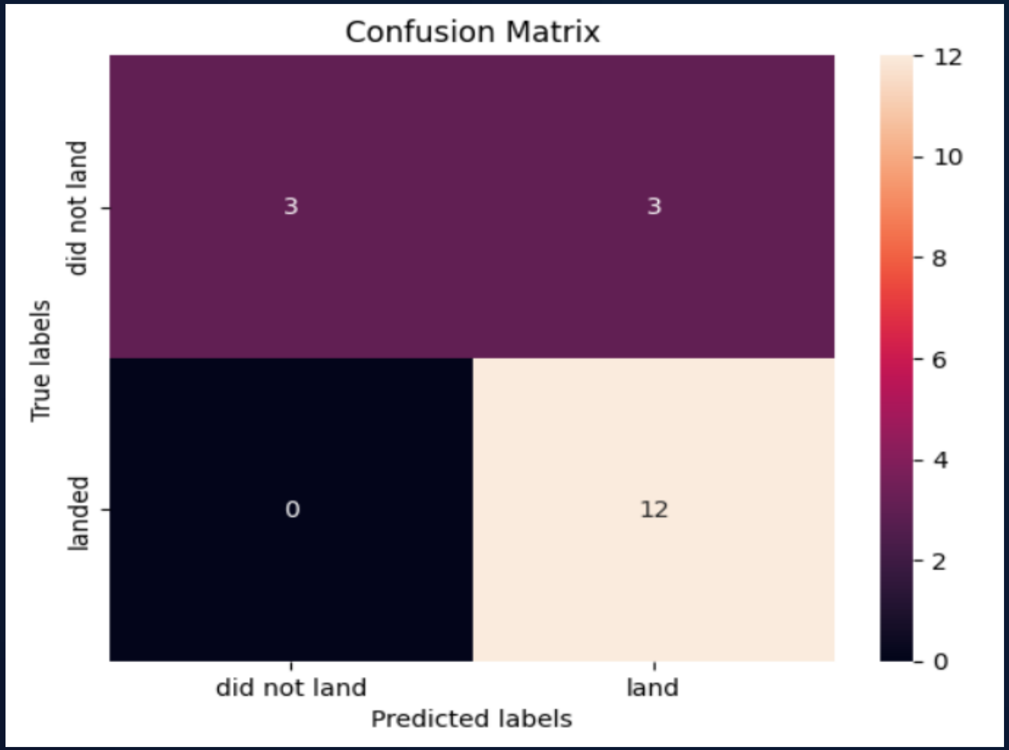


Machine Learning Prediction

Decision Tree

- GridSearchCV best score: 0.8892857142857142
- Accuracy score on test set: 0.8333333333333334

Confusion matrix:

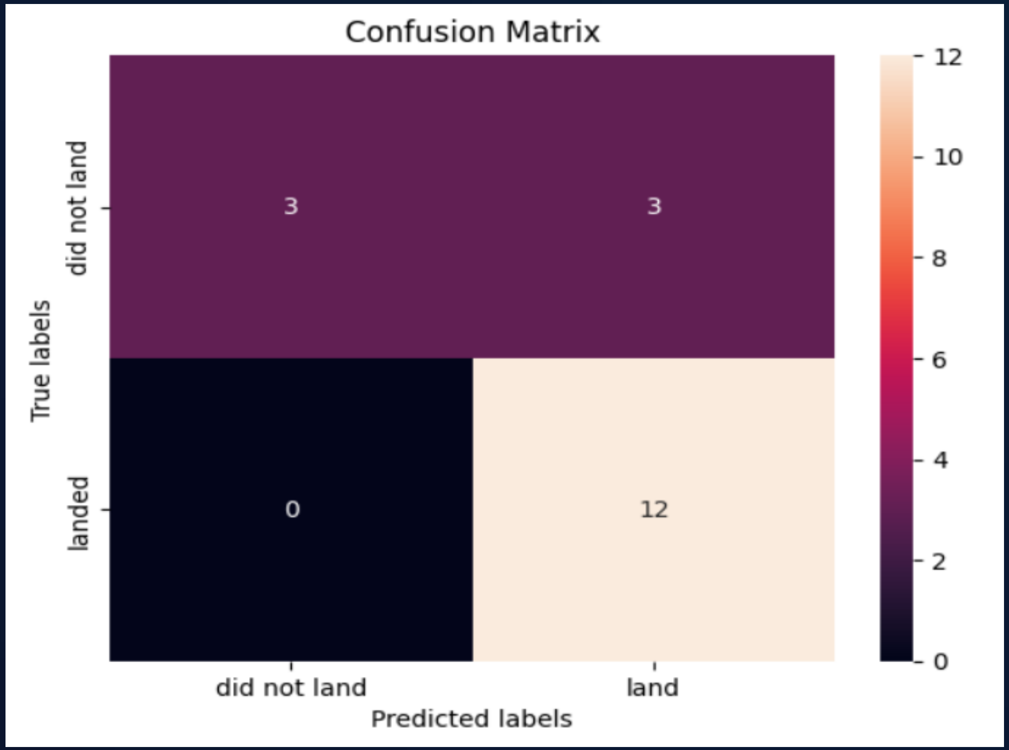


Machine Learning Prediction

K nearest neighbors (KNN)

- GridSearchCV best score: 0.8482142857142858
- Accuracy score on test set: 0.8333333333333334

Confusion matrix:



https://github.com/LahcenAmroune/Applied-Data-Science-Capstone/blob/main/8%20Machine%20Learning%20Prediction.ipynb

Discussion

Despite all four models—Logistic Regression, SVM, Decision Tree, and KNN—achieving identical test-set accuracy (83.3%) and confusion matrices, their performance diverged under hyperparameter tuning. To establish a definitive ranking, we compared their respective GridSearchCV best scores. Ordered from highest to lowest, the results are as follows: Decision Tree (0.889), K-Nearest Neighbors (0.848), Support Vector Machine (0.848), and Logistic Regression (0.846). This places the Decision Tree as the optimal model after parameter optimization.

Success rates have shown a steady upward trajectory over time across all factors, reflecting ongoing incremental operational improvements and technological advancements. Performance varies by orbit, with ES-L1, SSO, HEO, and GEO consistently achieving successful outcomes. Launch site also proved to be a highly predictive factor, with KSC LC-39A emerging as the top performer, closely followed by CCAFS LC-40. Among the predictive models evaluated, most achieved acceptable accuracy in forecasting landing outcomes, with the DecisionTreeClassifier delivering the best performance overall, demonstrating high accuracy, precision, and recall.

Appendix

Data sources

- r/SpaceX API Docs
- List of Falcon 9 and Falcon Heavy launches – Wikipedia

Conclusion

In this project, we set out to predict whether the first stage of a given Falcon 9 launch would land successfully, in order to determine the cost of a launch. Each feature of a Falcon 9 launch, such as payload mass or orbit type, may affect the mission outcome in a certain way. Several machine learning algorithms were employed to learn patterns from past Falcon 9 launch data and produce predictive models that can be used to predict the outcome of future launches. Among the four algorithms tested, the Decision Tree Classifier performed the best.

Our analysis revealed that success rates have increased over time across all factors, indicating continuous and incremental operational improvements and technological advancements. Different orbits exhibit varying success rates, with ES-L1, SSO, HEO, and GEO showing consistently successful outcomes. Launch site also proved to be a highly predictive factor, with KSC LC-39A being the top performer, closely followed by CCAFS LC-40.

Overall, many of the predictive models evaluated were able to predict landing outcomes with an acceptable level of accuracy. In the testing performed, the Decision Tree Classifier produced the best results, achieving high accuracy, precision, and recall.

If you'd prefer a shorter version, a bullet-point version, or a version that merges your original and new text differently, just let me know.

Thank you!

Presented by Lahcen Mokhtar AMROUNE • July 2026

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